

Problem 1: Information Sheet

MATH 301 Optimization: Project

May 6, 2026

1 Algorithm Description: Tailwind-Adaptive Adam

Our backbone idea for implementing `f_optimize` is based on the **Adam (Adaptive Moment Estimation)** [1]. Adam was chosen for its robustness in handling both sparse gradients and non-stationary objective functions by maintaining individual adaptive learning rates.

To improve performance, we enhanced the algorithm with a custom **"Tailwind" adaptive momentum** mechanism that dynamically scales the step size based on gradient alignment.

2 Mathematical Formulation

The iterative process utilizes exponentially decaying moving averages of the gradient (m) and its square (v).

2.1 Moment Updates

For each iteration t , given the gradient $g_t = \nabla f(x_t)$:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \text{ (First Moment/Momentum)} \quad (1)$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \text{ (Second Moment/Variance)} \quad (2)$$

2.2 Bias Correction

To prevent biased estimates during the initial steps :

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t} \quad (3)$$

2.3 Update Rule

The final parameter update is defined as :

$$x_{t+1} = x_t - \alpha_t \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \quad (4)$$

3 Algorithm Parameters

The following hyperparameters were utilized to ensure convergence across the varied test landscapes :

Parameter	Value	Description
α (Base)	0.05	Initial learning rate (step size)
α_{max}	0.50	Upper bound for Tailwind acceleration
β_1	0.9	Decay rate for the first moment
β_2	0.999	Decay rate for the second moment
ϵ	10^{-8}	Constant for numerical stability
Tol	10^{-6}	L2-norm gradient threshold for early stopping

Table 1: Hyperparameter settings for the optimization solver.

4 Innovative Ideas

The implementation incorporates two key innovations to fulfill the project’s requirement for original algorithmic improvements.

4.1 Alignment-Based Tailwind Acceleration

Instead of a fixed learning rate, the algorithm evaluates the *alignment* between the current gradient g_t and the existing momentum m_t :

- **Tailwind:** If $\langle g_t, m_t \rangle > 0$, the gradient aligns with the trajectory. The learning rate α increases by 5% per step to accelerate through flat regions or consistent slopes .
- **Headwind:** If $\langle g_t, m_t \rangle < 0$, the solver has overshoot the minimum. The momentum m is instantly reset to zero to prevent unstable oscillations, and α returns to the base value .

4.2 Historical Best State Tracking

Because iterative methods can oscillate or diverge, the function does not simply return the terminal coordinate x_T . The algorithm evaluates $f(x)$ at every iteration and stores the absolute **best approximate solution** found throughout the entire execution. This ensures the output satisfies the "best approximate solution" requirement even if the time limit expires during a divergent phase.

References

- [1] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” in *Proceedings of the 3rd International Conference on Learning Representations (ICLR)*, 2015.